Submitted for peer review

Research papers

Sediment flux follows water on the Tibetan Plateau but follows supply below its dams

Weixiang Zhao [a,b], Huilin Wang [b,*]

[a] Dongguan Jiarong Foreign Language School, Dongguan, Guangdong 523145, China
[b] College of Water Conservancy and Civil Engineering, South China Agricultural University, Guangzhou, Guangdong 510642, China

ARTICLE INFO

Keywords:
Tibetan Plateau
Sediment flux
Climate elasticity
Sediment rating curve
Transport capacity
Channel storage
Three Gorges Dam

ABSTRACT

Suspended-sediment flux from Tibetan Plateau rivers has increased with regional warming, an increase that has been attributed to sediment released by retreating glaciers and thawing permafrost. We analyse annual records with a reach-scale mass balance between sediment supply, transport capacity and channel storage. In a sediment-rich reach the balance predicts that outflow is set by capacity and does not respond to inflow; in a sediment-starved reach outflow follows inflow one-for-one. We test the second prediction in the Jingjiang reach of the middle Yangtze, below the Three Gorges Dam, where the inflow transmission coefficient is $\lambda = 0.96 \pm 0.09$ (predicted value 1) and the bed store decays within about a decade where the bed armours. In 16 plateau rivers, sediment flux scales with roughly the 2.3 power of runoff, and its response to precipitation passes almost entirely through runoff. In the only gauged plateau reach, on the upper Indus, $\lambda = 0.34 \pm 0.17$, lower than below the dams ($\Delta\lambda = 0.66$, 95% interval 0.29–1.07) but weakly constrained. Despite warming of 0.1–0.6 °C per decade, no direct effect of temperature on sediment flux at fixed runoff is detectable, and the detectable temperature-associated changes are mediated by runoff. With a rating exponent of 2.29, a runoff increase of 5% per decade implies about 12% per decade in sediment flux, sufficient to explain the regional increase previously attributed to sediment release. The results are consistent with predominantly capacity-controlled sediment export from the plateau, although slower changes in sediment supply remain unresolved.

1. Introduction

Rivers draining steep, tectonically active mountains deliver a disproportionate share of the sediment that reaches the oceans (Milliman and Syvitski, 1992; Syvitski and Milliman, 2007). This sediment builds floodplains and deltas, fills reservoirs, carries nutrients and particulate organic carbon, and controls the stability of channels used for navigation, irrigation and flood defence (Walling, 2006; Kondolf et al., 2014). The Tibetan Plateau and its bordering ranges supply the Yangtze, Yellow, Mekong, Salween, Brahmaputra, Ganges and Indus, as well as the endorheic Tarim and Qaidam basins, on which more than a billion people depend for water, food and energy (Immerzeel et al., 2010). The plateau has warmed at about twice the global mean rate over the past half-century (Kuang and Jiao, 2016), glaciers have retreated almost everywhere except in the Karakoram (Yao et al., 2012), and permafrost has warmed and thinned. The hydrological response, a near-term rise in glacier melt followed by a decline as glaciers shrink, has been studied extensively (Immerzeel et al., 2010). The sediment response is less well documented, although sediment loads control reservoir sedimentation and the stability of channels downstream (Kondolf et al., 2014).

Several recent studies report rising sediment export from the plateau. Li et al. (2021) compiled six decades of records from 28 quasi-pristine headwater basins of High Mountain Asia and reported that sediment flux

* Corresponding author.
E-mail address: whl@scau.edu.cn (H. Wang).

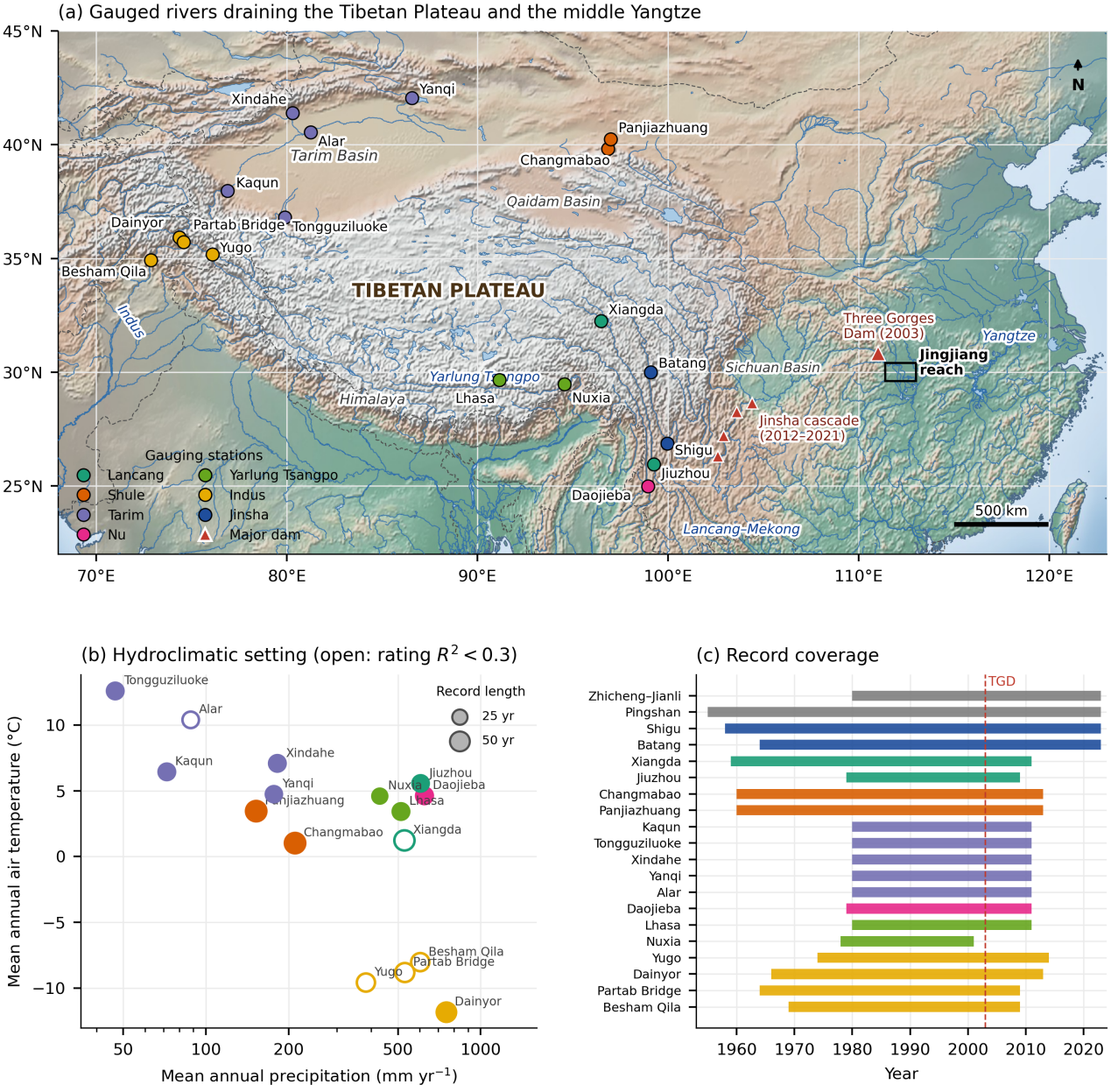


Fig. 1. Study domain. (a) Gauging stations on rivers draining the Tibetan Plateau, coloured by basin, on shaded relief; triangles mark the Jinsha River cascade (Wudongde, Baihetan, Xiluodu and Xiangjiaba, impounded 2012–2021) and the Three Gorges Dam (2003), and the box marks the Jingjiang reach of the middle Yangtze. Station locations are approximate. (b) Hydroclimatic setting of the plateau stations: record-mean annual precipitation and air temperature, with symbol size proportional to record length; open symbols denote stations with poorly defined sediment ratings ($R^2 < 0.3$). The southeastern, monsoon-fed basins (Lancang, Nu, Yarlung Tsangpo) are wet and mild, the northern, glacier-fed Tarim and Shule basins are arid, and the upper Indus is cold. (c) Periods covered by the annual runoff and sediment records used in this study; the dashed line marks the impoundment of the Three Gorges Dam. Coastline, lakes, borders and rivers from GSHHG/WDB-II (intermediate resolution); relief from the Natural Earth shaded-relief raster distributed with Basemap.

increased by about 12% per decade, compared with about 5% per decade for runoff. Using a climate elasticity model, they found that the sensitivity of sediment flux to precipitation decreases, and its sensitivity to temperature increases, with glacier cover. They attributed the regional increase to a warmer and wetter climate that releases sediment from glacier forefields and thawing permafrost and intensifies erosive rainfall, and projected that sediment flux could more than double by 2050 under high emissions. Across 151 rivers, Li et al. (2024a) identified glacier cover as the first-order spatial control on suspended sediment yield, with precipitation amplifying its effect nonlinearly and vegetation acting in both directions. In four cryosphere-fed basins of the Tien Shan and northern plateau, Zhang et al. (2023) documented seasonal shifts from snowmelt- to rainfall- and glacier-melt-dominated transport and reproduced them with a monthly sediment availability–transport model. In the Tuotuo River, the source of the Yangtze, Zhang et al. (2020) found a stepwise increase in sediment flux after 1997 and attributed it mainly to the greater erosive power and transport capacity of increased discharge rather than to new sediment sources. Satellite reconstructions indicate that about 30% of the suspended sediment mobilised in plateau headwaters between 1986 and 2021 was deposited temporarily within river channels (Li et al., 2024b).

These studies do not resolve whether rising sediment flux reflects more available sediment, for example from deglaciated moraines, paraglacial slopes and thawing permafrost (Hallet et al., 1996; Ballantyne, 2002), or more water to transport sediment that was already present. The two mechanisms imply different futures. A supply-driven increase can outlast its climatic cause, because newly exposed sediment remains erodible for decades to millennia (Ballantyne, 2002). A transport-driven increase scales with runoff and would reverse if runoff declined, for instance after peak glacier melt (Immerzeel et al., 2010). Total climate elasticities cannot separate the two, because the same climatic drivers act on both runoff and sediment availability. Consider a basin in which a 1% increase in precipitation raises sediment flux by 3%. This could reflect a 1.3% increase in runoff acting on a rating with exponent 2.3, with no change in availability, or a 1% increase in runoff acting on a rating with exponent 1 together with a 2% increase in availability. Both cases give the same total elasticity, but only the second involves a change in sediment availability.

The distinction between transport and supply is long established in fluvial geomorphology. A river is transport-limited when it carries as much sediment as its flow can move, and supply-limited when it carries less because sediment is not available (Einstein, 1950; Church, 2006). Einstein (1950) divided the suspended load into a bed-material load, whose flux is set by local hydraulics, and a finer wash load, whose flux is set by upstream supply. Lane (1955) stated the balance qualitatively: a channel aggrades when sediment supply exceeds the product of discharge and slope and degrades when it falls short. Its quantitative form is the Exner equation of sediment continuity (Exner, 1925; Paola and Voller, 2005), which, integrated over a reach, relates the change in stored sediment to the difference between inflow and outflow.

The most widely observed expression of this balance is the sediment rating relation,

$$S = aQ^b, \tag{1}$$

which relates suspended-sediment flux S to water discharge Q (Walling, 1977; Asselman, 2000; Syvitski et al., 2000; Horowitz, 2003). The coefficient a is commonly interpreted as an index of sediment availability, and the exponent b as the rate at which transport increases with flow (Asselman, 2000; Syvitski et al., 2000). Where sediment is abundant, concentration rises steeply with discharge and b approaches the exponent of transport capacity, typically about 1.5–3 (Leopold and Maddock, 1953; Syvitski et al., 2000). Where sediment is scarce, concentration is set by supply and may be diluted at high flow, so that $b \lesssim 1$. Departures from a fixed rating at constant discharge indicate changes in availability (Topping et al., 2000; Rubin and Topping, 2001; Warrick, 2015). The climate elasticity of streamflow, the proportional change in runoff per proportional change in a climatic variable, was introduced by Schaake (1990) and estimated nonparametrically by Sankarasubramanian et al. (2001). Combined with the rating, it leads to a chain rule in which the climate sensitivity of sediment flux is split into a part mediated by runoff and a direct part acting on availability at fixed flow, as in path analysis (Baron and Kenny, 1986). We are not aware of a systematic application of this decomposition to mountain rivers.

An absence of a direct climate effect could indicate a capacity-controlled river or simply a lack of statistical power. Interpreting plateau records in these terms therefore requires a test in a river where the supply–capacity balance is known independently. The Yangtze provides such a case at the downstream end of the plateau's own sediment system. Its sediment flux to the sea has fallen by about two-thirds since the 1960s (Yang et al., 2011, 2015; Dai and Lu, 2014; Guo et al., 2019). The Three Gorges Dam (TGD), impounded in 2003, traps most of the sediment entering its reservoir (Yang et al., 2014), and since 2012 the Jinsha River cascade has intercepted nearly all of the sediment leaving the plateau. The middle Yangtze has eroded its bed continuously since impoundment (Xia et al., 2016; Lai et al., 2017; Lyu et al., 2019). The coarse fraction of the suspended load recovered at Jianli to roughly pre-dam levels in the first years after impoundment, but the recovery weakened as the bed coarsened, and the fine fraction did not recover (Zhang et al., 2017; Yang et al., 2022). Lai et al. (2017) argued that scour will stop only when the bed becomes too coarse to move and that the locus of scour has migrated downstream. In this reach, supply was cut abruptly and independently of climate, and the response of the bed can be measured from sediment budgets.

This study examines whether the rising sediment export of the Tibetan Plateau reflects more available sediment or more water. We formulate a reach-scale balance between supply, capacity and storage that permits both erosion and deposition, derive its predictions in the sediment-rich and sediment-starved limits, and estimate the rating exponent expected under capacity control from transport theory and hydraulic geometry. We then address four questions. (i) Does the balance predict how the Jingjiang reach below the TGD responds to discharge and supply? (ii) Are the ratings of plateau rivers consistent with capacity control? (iii) How much of the climate sensitivity of plateau sediment flux passes through runoff, and how much reflects a direct effect of climate on availability at fixed flow? (iv) Where has availability changed, and what does the balance imply for sediment released by warming? The analysis uses annual records from 16 plateau stations, two upper Jinsha River stations and the Jingjiang reach of the middle Yangtze.

2. Theory

2.1. Mass balance

In one dimension, conservation of sediment mass in a channel is expressed by the Exner equation,

$$(1-\lambda_p)\,\frac{\partial \eta}{\partial t} = -\frac{\partial q_s}{\partial x}, \tag{2}$$

where η is bed elevation, λ_p bed porosity, q_s the volumetric sediment flux per unit width, x the downstream coordinate and t time (Exner, 1925; Paola and Voller, 2005). Integrating Eq. (2) over a reach of length L and width B, with fluxes expressed as mass rates, gives

$$S_{out} = S_{in} + J, \tag{3a}$$

$$\frac{dM}{dt} = -J, \tag{3b}$$

where $S_{in} = S_{up} + S_{lat}$ is the net sediment input to the control volume, the upstream load S_{up} plus the signed net lateral input S_{lat} (positive for tributary inflow, negative for diversions), S_{out} is the load leaving the reach at its downstream boundary, M is the mass of exchangeable sediment stored in the bed and banks, and J is the net bed-exchange flux, positive for erosion and negative for deposition. Equation (3) is exact; it becomes a model only when J is specified.

2.2. Closure

We close Eq. (3) by assuming that the outflow cannot exceed the transport capacity of the flow, $S_c(Q)$, and that the bed cannot release sediment faster than a rate set by the remaining store:

$$J = \begin{cases} \min\left[k_e M^{\gamma} f(Q),\; S_c - S_{in}\right], & S_{in} < S_c, \\ -\left(S_{in} - S_c\right), & S_{in} \geq S_c, \end{cases} \tag{4}$$

where the upper branch applies to a sediment-starved reach and the lower branch to a sediment-rich reach, k_e is an erodibility coefficient, γ an exponent describing how the release rate depends on the store, and $f(Q) = (Q/\bar{Q})^m$ a hydraulic forcing with exponent m. The upper branch is a reach-scale analogue of the active-layer and armouring concepts of Hirano (1971) and Parker and Klingeman (1982): as erodible sediment is removed, the bed coarsens and the release rate declines. For $\gamma = 1$ and constant forcing it reduces to the rate law of fluvial adjustment, $M = M_0 e^{-k_e t}$ (Graf, 1977; Wu et al., 2012). The lower branch states that inflow in excess of capacity is deposited within the reach in the same year. Equation (4) is an idealised closure, not a consequence of the Exner equation: it assumes that deposition adjusts fully to capacity within one year and neglects travel times within the reach and changes in capacity caused by the deposition itself. Dimensional consistency requires $[k_e] = \mathrm{Mt}^{1-\gamma}\,\mathrm{yr}^{-1}$, which reduces to yr^{-1} only for $\gamma = 1$. The supply ratio

$$\Pi_s = \frac{S_{in}}{S_c} \tag{5}$$

classifies the regime: $\Pi_s \geq 1$ is sediment-rich and $\Pi_s < 1$ sediment-starved.

Transport capacity is defined for sediment that is represented in the bed. The measured suspended load also contains wash load, whose flux is set mainly by upstream supply (Einstein, 1950). Writing the load as $S = S_b + S_w$, where S_b is the bed-material component and S_w the wash load, the capacity constraint applies to the first component,

$$S_{b,out} = \min\left(S_{b,in} + E_b,\ S_{c,b}\right), \tag{6}$$

whereas the wash load is governed by a budget without a capacity limit,

$$S_{w,out} \approx S_{w,in} + E_w - D_w, \tag{7}$$

where E_b and E_w are bed releases of the two components and D_w is deposition of fines. The supply–capacity balance, Eq. (4), is strictly applicable to the bed-material component. Because grain-size-partitioned loads are not available for most stations, it is applied here to the total suspended load as an effective reach-scale approximation (Section 2.7). Under this approximation, an observed rating exponent reflects both components, and its interpretation as a capacity exponent is valid only where bed-material load dominates the annual flux. The simplifications involved are listed in Section 2.7.

2.3. Two limits

Sediment-rich limit ($\Pi_s \geq 1$). The outflow equals capacity, $S_{out} = S_c(Q)$, so that

$$\frac{\partial \ln S_{out}}{\partial \ln Q} = \frac{d \ln S_c}{d \ln Q} \equiv b_c, \tag{8a}$$

$$\frac{\partial \ln S_{out}}{\partial \ln S_{in}} = 0. \tag{8b}$$

Sediment flux follows water, with a rating exponent equal to the exponent of capacity, and does not respond to changes in supply. Additional sediment delivered to the reach is deposited ($J < 0$).

Sediment-starved limit ($\Pi_s < 1$, storage-limited release). The outflow equals supply plus bed release, $S_{out} = S_{in} + k_e M^\gamma f(Q)$. If bed release does not depend on S_{in},

$$\frac{\partial \ln S_{out}}{\partial \ln S_{in}} = \frac{S_{in}}{S_{out}}, \tag{9a}$$

$$\frac{\partial \ln S_{out}}{\partial \ln Q} = m\left(1 - \frac{S_{in}}{S_{out}}\right). \tag{9b}$$

Sediment flux follows supply, with an elasticity equal to the share of supply in the outflow. The discharge derivative in Eq. (9b) is evaluated at fixed sediment inflow and bed storage. Along an actual hydrological path, inflow also varies with discharge, and at fixed M the total response is

$$\frac{d \ln S_{out}}{d \ln Q} = \frac{S_{in}}{S_{out}} \frac{d \ln S_{in}}{d \ln Q} + m\left(1 - \frac{S_{in}}{S_{out}}\right), \tag{10}$$

with an additional term $\gamma\,(1 - S_{in}/S_{out})\,d\ln M/d\ln Q$ if the store itself depends on discharge. A small partial elasticity $\partial \ln S_{out}/\partial \ln Q$ therefore does not imply that outflow is insensitive to discharge along the path $Q \to S_{in} \to S_{out}$. Bed release declines as M is depleted, at a rate set by k_e and γ.

Equation (9a) is a local derivative at fixed discharge and bed state. Because S_{in}/S_{out} varies from year to year, a regression slope across years estimates an average of this quantity whose value depends on how the average is taken. A more direct test uses the structural form of the starved limit,

$$S_{out} = \lambda\, S_{in} + B_0 \left(\frac{Q}{\bar{Q}}\right)^{m} e^{-t/\tau}, \tag{11}$$

which combines the upper branch of Eq. (4) with $\gamma = 1$. Mass conservation with separable bed release (Section 2.7) requires $\lambda = 1$, so that each additional tonne of inflow appears as one additional tonne of outflow. In the sediment-rich limit S_{out} is independent of S_{in} and $\lambda = 0$. Intermediate values can arise from partial capacity limitation, temporary storage within the reach, grain-size-selective deposition, travel times longer than a year, or a mixture of sub-reaches in different regimes. When estimated from annual data, λ is a marginal transmission coefficient, the fraction of interannual variations in inflow that appears in the outflow within the same year; it is not the fraction of the sediment mass that passes through the reach.

For $\gamma = 1$ and time-varying forcing, the store obeys $dM/dt = -k_e M f(Q)$, which integrates to

$$M(t) = M_0 \exp\left[-k_e \int_0^t f\big(Q(s)\big)\, ds\right], \tag{12}$$

so that bed release decays with cumulative hydraulic forcing rather than with calendar time. The calendar-time form in Eq. (11) is the special case of constant forcing and is used as an approximation; both forms are fitted (Section 4).

2.4. Capacity exponent

The value of b_c in the sediment-rich limit can be estimated from transport theory. The total-load formula of Engelund and Hansen (1967) can be written

$$q_s \propto U^2\,(hS_f)^{3/2}\left[(s-1)^2 g^{1/2} d_{50}\right]^{-1}, \tag{13}$$

where U is mean velocity, h flow depth, S_f energy slope, s the specific gravity of sediment, g gravitational acceleration and d_{50} the median grain size. With at-a-station hydraulic geometry, $B \propto Q^{\beta_w}$, $h \propto Q^{\beta_h}$ and $U \propto Q^{\beta_u}$ with $\beta_w + \beta_h + \beta_u = 1$ (Leopold and Maddock, 1953), and constant slope and grain size, the instantaneous capacity exponent is

$$b_c^* = \beta_w + 2\beta_u + \tfrac{3}{2}\beta_h. \tag{14}$$

For the average at-a-station exponents of Leopold and Maddock (1953) ($\beta_w \approx 0.26$, $\beta_h \approx 0.40$, $\beta_u \approx 0.34$), $b_c^* \approx 1.5$. The annual load is $S = \int_{\mathrm{yr}} S_c(Q(t))\,dt$. Writing $Q(t) = \bar{Q}\,\phi(t)$, where $\bar{Q}$ is the annual mean and ϕ a normalized hydrograph, gives $S \propto \bar{Q}^{\,b_c^*}\,\langle \phi^{\,b_c^*}\rangle$, so that the annual rating exponent is

$$b = b_c^* + \frac{d\ln\langle \phi^{\,b_c^*}\rangle}{d\ln \bar{Q}}. \tag{15}$$

The annual exponent exceeds b_c^* when wet years also have more peaked hydrographs, as is typical of monsoon and melt-driven regimes. Two caveats apply. Equation (13) describes bed-material transport in sand-bed channels, and its use for the total suspended load, which includes wash load, is an approximation (Section 2.7). In addition, an annual exponent of about 2.3 with $b_c^* \approx 1.5$ requires the hydrograph term in Eq. (15) to contribute about 0.75, which could not be verified without daily discharge data. The theory thus indicates that values of b well above 1 are expected under capacity control but does not predict the observed value. A steep rating is also not sufficient for capacity control, because supply can increase nonlinearly with discharge, for example through rainfall-triggered hillslope erosion or bank failure (Asselman, 2000; Syvitski et al., 2000). Additional diagnostics are described in Section 4.

2.5. Climate pathways

The annual sediment flux at a station is written as

$$\ln S = \ln a + b \ln Q + c_T T' + c_P \ln P' + \epsilon_S, \tag{16}$$

where $T' = T - \bar{T}$ is the temperature anomaly, $P' = P/\bar{P}$ normalized precipitation, and c_T and c_P the direct effects of temperature and precipitation on sediment flux at fixed runoff. Equation (16) extends Eq. (1) by allowing the availability term a to vary with climate. Runoff depends on climate as

$$\ln Q = \alpha + \frac{\partial \ln Q}{\partial T} T' + \varepsilon_P^Q \ln P' + \epsilon_Q, \tag{17}$$

where ε_P^Q is the precipitation elasticity of runoff (Schaake, 1990; Sankarasubramanian et al., 2001) and $\partial \ln Q / \partial T$ the proportional change in runoff per degree of warming. Substituting Eq. (17) into Eq. (16) gives the total climate sensitivities of sediment flux:

$$\varepsilon_P^S = \underbrace{b\,\varepsilon_P^Q}_{\text{runoff pathway}} + \underbrace{c_P}_{\text{direct pathway}}, \tag{18a}$$

$$\frac{\partial \ln S}{\partial T} = \underbrace{b \frac{\partial \ln Q}{\partial T}}_{\text{runoff pathway}} + \underbrace{c_T}_{\text{direct pathway}}. \tag{18b}$$

The runoff pathway represents more water carrying sediment that is already available; the direct pathway represents changes in availability at a given flow. In path-analysis terms (Baron and Kenny, 1986), these are the effect mediated by runoff and the direct effect.

In the sediment-rich limit, Eq. (8) implies $c_T \approx c_P \approx 0$, because additional sediment is stored rather than exported, and $b \approx b_c > 1$, so that changes in runoff are amplified. In the sediment-starved limit, sediment flux is insensitive to runoff, and c_T or c_P can be large if climate changes supply. Equation (18) decomposes statistical associations. A direct coefficient c_T indistinguishable from zero means that no temperature effect at fixed runoff is detectable in interannual variability; it does not mean that warming has no effect on sediment supply. Slow, cumulative effects of warming and effects that covary with runoff are not identified by c_T and are examined with separate diagnostics. For a long-term trend, and provided that the interannual exponent also applies to long-term change, proportional rates of change of sediment flux, r_S, and runoff, r_Q, over a fixed interval without change in availability are related by

$$1 + r_S = (1 + r_Q)^b. \tag{19}$$

2.6. Availability diagnostics

Two diagnostics address slow changes in availability that the interannual decomposition may not resolve. The discharge-conditioned anomaly

$$I_s(t) = \ln S(t) - [a_0 + b_0 \ln Q(t)], \tag{20}$$

with (a_0, b_0) fitted to a baseline period, measures the change in sediment flux at constant discharge, following Topping et al. (2000) and Rubin and Topping (2001). A positive trend in I_s indicates increasing availability and a negative trend depletion. A change in rating shape, $b_1 \to b_2$, records availability changes that depend on flow: steepening indicates that high flows access additional sediment, and flattening that high flows become relatively sediment-poor, as expected for source exhaustion or dilution by meltwater (Asselman, 2000; Warrick, 2015).

2.7. Assumptions

The analysis rests on the following assumptions, which are revisited in Section 6.

- *Reach averaging.* Each reach is treated as a single well-mixed control volume; the downstream propagation of disturbances within a reach, and the spatial structure of erosion and deposition, are not resolved. Sub-reach budgets are used where gauges allow.

- *Annual time scale.* All quantities are annual totals or means. The capacity S_c is the capacity integrated over the annual hydrograph (Section 2.4); intra-annual hysteresis, flood sequencing and seasonal exhaustion are not represented.

- *Single effective fraction.* The capacity constraint applies to the bed-material component, Eq. (6), whereas wash load follows Eq. (7) (Einstein, 1950). Without grain-size-partitioned loads, the balance is applied to total suspended load as an effective approximation.

- *Separability of bed release.* In the starved limit, bed release $k_e M^{\gamma} f(Q)$ is assumed not to depend on S_{in}. This yields $\lambda = 1$ in Eq. (11) and the elasticity of Eq. (9a); a correlation between bed release and inflow at fixed discharge would bias the estimated supply elasticity upward.

- *Log-linear climate response.* Equations (16)–(17) assume that log sediment flux and log runoff respond linearly to temperature anomalies and to log precipitation, with constant coefficients over each record. Changes in the coefficients are tested separately through rating-shape changes.

- *Stationary basin conditions.* Apart from the impoundments considered explicitly, land use, diversions and regulation are assumed not to change systematically during each record. Stations where this is doubtful are identified in Section 3.

- *Measurement error.* Annual loads are assumed to carry independent, proportional errors of a few percent (Appendix A); errors common to several gauges, such as those arising from shared sampling methods, are not modelled.

- *Interannual and long-term elasticities.* Using Eq. (19) for decadal trends assumes that the rating exponent estimated from interannual variability also applies to slow, sustained changes in runoff. This is tested by comparing exponents estimated from year-to-year differences and from multi-year means.

3. Data

3.1. Plateau stations

The 16 headwater stations drain the northern, eastern, southern and western margins of the plateau (Fig. 1a; Table 4) and span climates from the warm, wet monsoon basins of the southeast to the arid Tarim and Shule basins and the cold upper Indus (Fig. 1b). Record periods are shown in Fig. 1c. The Lancang (upper Mekong; Xiangda and Jiuzhou), Nu (Salween; Daojieba) and Yarlung Tsangpo (upper Brahmaputra; Lhasa on the Lhasa River tributary and Nuxia on the main stem) drain the southeastern plateau, where most precipitation falls during the summer monsoon. The Shule River (Changmabao and Panjiazhuang) and the Tarim tributaries (Kaqun on the Yarkand, Tongguziluoke on the Yurungkash, Xindahe in the Aksu system, Yanqi on the Kaidu, and Alar on the Tarim main stem) drain glaciated ranges of the Qilian Shan, Kunlun, Karakoram and Tien Shan into arid basins, where runoff depends strongly on glacier and snow melt. The upper Indus and its tributaries (Yugo on the Shyok, Dainyor on the Hunza–Gilgit system, Partab Bridge and Besham Qila on the main stem) drain the Karakoram and western Himalaya, where westerly precipitation and glacier melt dominate. Lowland gauges on the Tarim (Alar) and Shule (Panjiazhuang) are affected by irrigation diversions and reservoirs (Section 2.7).

For each station, annual runoff (Q, 10^8 m^3 yr^{-1}), annual suspended-sediment load (S, 10^4 t yr^{-1}), basin-mean annual air temperature (T, °C) and annual precipitation (P, mm yr^{-1}) were available for 23–53 years between 1959 and 2013. The stations correspond to the headwater dataset of Li et al. (2021). Loads at the four upper Indus stations are one to three orders of magnitude smaller than at the other stations, which suggests that they are reported in different units; because all analyses use logarithms within stations, this affects intercepts

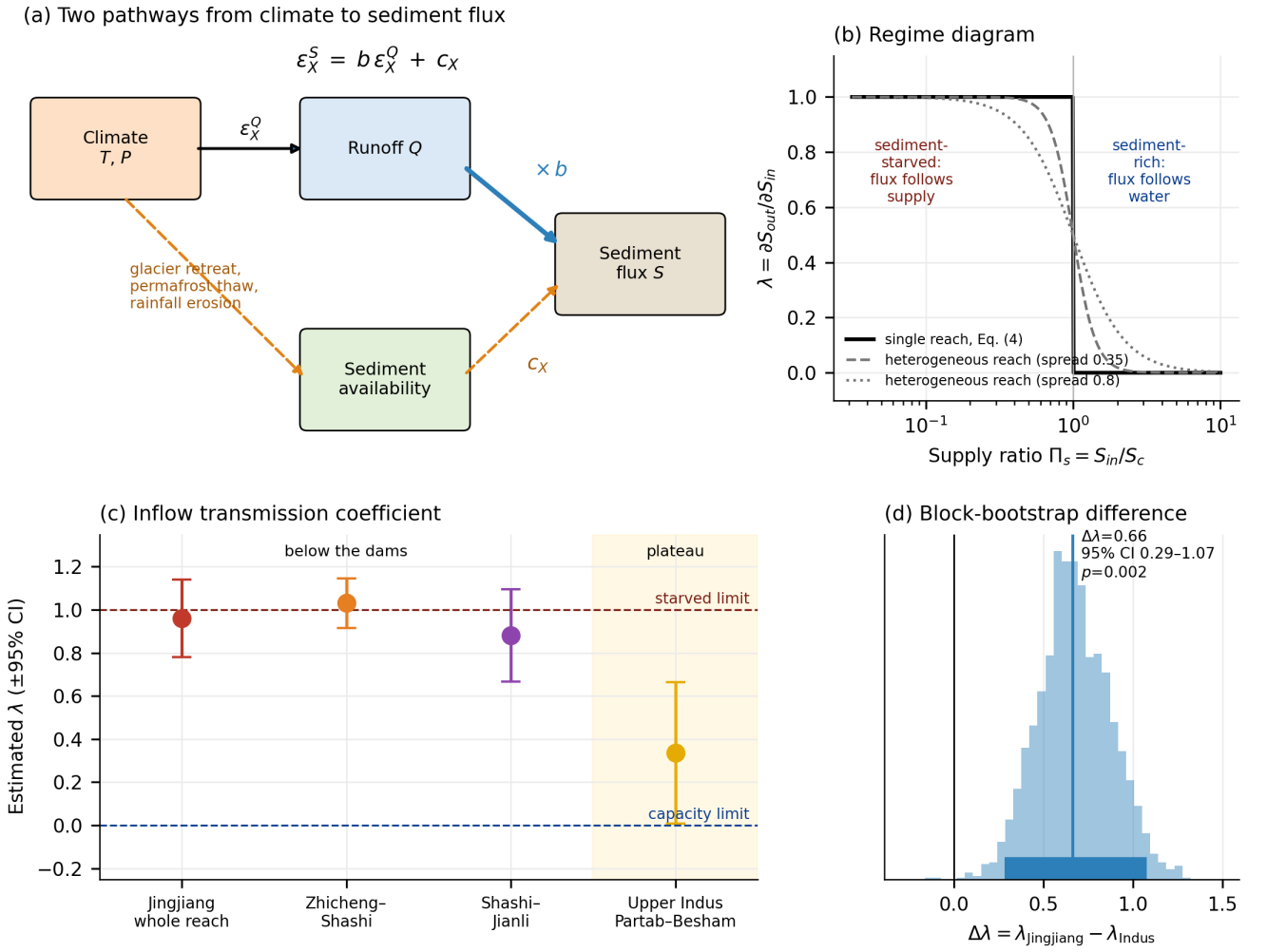


Fig. 2. (a) Two pathways from climate to sediment flux (Eq. 18): through runoff, amplified by the rating exponent b, and directly through sediment availability (c_X). (b) Regime diagram: the inflow transmission coefficient $\lambda = \partial S_{out}/\partial S_{in}$ as a function of the supply ratio Π_s for a single reach (Eq. 4, step) and for heterogeneous reaches in which Π_s varies between sub-reaches or years (smoothed curves, illustrative). (c) Estimated λ (Eq. 11) with 95% confidence intervals for the Jingjiang reach and sub-reaches below the dams and for the upper Indus on the plateau. (d) Block-bootstrap distribution of the difference $\Delta\lambda$ between the Jingjiang reach and the upper Indus. The observed λ are effective coefficients estimated from interannual variability and are not instantaneous derivatives.

but not exponents, elasticities or the Indus reach test. Two stations on the upper Jinsha River, Batang and Shigu (1958–2022), represent the plateau outlet of the Yangtze; runoff and sediment load are available for them, but climate series are not.

3.2. Yangtze stations

Annual runoff and sediment load for 1955–2022 at Pingshan (the Jinsha outlet below the cascade), Cuntan (inflow to the TGD reservoir), Yichang (below the TGD), and stations in the Jingjiang reach were compiled from the *China River Sediment Bulletin* (Ministry of Water Resources of China, 2022). The Jingjiang reach, between Zhicheng and Chenglingji, crosses the Jianghan Plain and loses water and sediment to Dongting Lake through three outlets: Songzi (gauged at Xinjiangkou and Shadaoguan), Taiping (Mituosi) and Ouchi (Kangjiagang and Guanjiapu). We analyse the gauged reach from Zhicheng to Jianli and its two sub-reaches separated by Shashi. The upper sub-reach (Zhicheng–Shashi) has a sand–gravel bed; the lower sub-reach (Shashi–Jianli) is a meandering sand-bed channel (Xia et al., 2016; Lyu et al., 2019). Upstream impoundments were: TGD, 2003; Xiangjiaba and Xiluodu, 2012–2013; Wudongde, 2020; Baihetan, 2021. Surveyed bankfull-channel erosion of the Jingjiang reach for October 2002 to October 2020, 12.29×10^8 m^3, was taken from Ministry of Water

Resources of China (2020).

Table 1. Principal variables and parameters.

Symbol	Definition	Unit
Q, $\bar{Q}$	Annual runoff; its mean	10^8 m^3 yr^{-1}
S, S_{in}, S_{out}	Annual suspended-sediment load; net input to and output from a reach	Mt yr^{-1}
S_{up}, S_{lat}	Upstream load; signed net lateral input	Mt yr^{-1}
S_b, S_w	Bed-material and wash-load components of S	Mt yr^{-1}
E_b, E_w, D_w	Bed release of the two components; deposition of fines	Mt yr^{-1}
T, T'	Basin-mean annual air temperature; anomaly from station mean	°C
P, P'	Annual precipitation; normalized by station mean	mm yr^{-1}, –
a, b	Coefficient and exponent of the rating $S = aQ^b$	–, –
b_c, b_c^*	Annual and instantaneous exponents of transport capacity	–
β_w, β_h, β_u	Hydraulic-geometry exponents of width, depth and velocity	–
ε_P^Q	Precipitation elasticity of runoff	–
$\partial \ln Q/\partial T$	Temperature sensitivity of runoff	°C^{-1}
ε_P^S, $\partial \ln S/\partial T$	Total precipitation elasticity and temperature sensitivity of sediment flux	–, °C^{-1}
c_P, c_T	Direct (fixed-Q) effects of precipitation and temperature on sediment flux	–, °C^{-1}
I_s	Discharge-conditioned sediment anomaly, Eq. (20)	–
b_1, b_2	Rating exponent in the first and second halves of a record	–
J	Net bed-exchange flux of a reach, $S_{out} - S_{in}$	Mt yr^{-1}
M, M_0	Exchangeable sediment store; initial value	Mt
k_e, γ, m	Erodibility coefficient, storage exponent, discharge exponent of bed release	Mt$^{1-\gamma}$ yr^{-1}, –, –
Π_s	Supply ratio S_{in}/S_c	–
β_Q, β_S	Elasticities of S_{out} with respect to Q and S_{in}	–, –
λ, B_0	Inflow coefficient and reference bed release in Eq. (11)	–, Mt yr^{-1}
b_{-1}	Coefficient of previous-year runoff in the exhaustion test	–
MDE	Minimum detectable effect, 2.8 SE	°C^{-1}
τ	e-folding time of post-dam bed erosion, $1/k_e$ for $\gamma = 1$	yr
η, λ_p, q_s	Bed elevation, porosity, volumetric sediment flux per unit width	m, –, m^2 s^{-1}
U, h, B, S_f, d_{50}	Velocity, depth, width, energy slope, median grain size	m s^{-1}, m, m, –, mm

4. Methods

4.1. Starved-reach test

Annual bed exchange in the Jingjiang reach was computed from Eq. (3) as

$$J = S_{\mathrm{JL}} + S_{\mathrm{XJK}} + S_{\mathrm{SDG}} + S_{\mathrm{MTS}} + S_{\mathrm{KJG}} + S_{\mathrm{GJP}} - S_{\mathrm{ZC}}, \tag{21}$$

where subscripts denote Jianli, the five outlet gauges and Zhicheng; analogous budgets were formed for the two sub-reaches (Appendix A). Equation (9) was tested by fitting

$$\ln S_{out} = \beta_0 + \beta_Q \ln Q + \beta_S \ln S_{in} + \epsilon \tag{22}$$

for 1980–2002 and 2004–2022 separately and comparing β_S with the observed share of inflow in outflow, summarised as the ratio of means, $\bar{S}_{in}/\bar{S}_{out}$, the mean of annual ratios and their median. Because bed release declines with time, Eq. (22) was also fitted with a linear time trend. Equation (11) was fitted by nonlinear least squares, and λ was compared with the starved-limit value of 1 and the sediment-rich value of 0. Collinearity between $\ln Q$ and $\ln S_{in}$ was quantified by their correlation. Equation (11) was also fitted in the form implied by Eq. (12), $S_{out} = \lambda S_{in} + B_0\, f(Q) \exp[-k_e F(t)]$, with $f(Q) = (Q/\bar{Q})^m$ and cumulative forcing $F(t) = \sum_{s=1}^{t} f(Q_s) - \frac{1}{2} f(Q_t)$, the discrete, mid-year form of $\int_0^t f\, ds$. The same exponent m enters $f(Q)$ and $F(t)$ and is estimated jointly with λ, B_0 and k_e; $1/k_e$ is reported as a decay scale in mean-flow years. The two forms were compared by AICc. The decline of bed release was described by the rate law with discharge forcing, $\ln J = a_J + m \ln(Q/\bar{Q}) - t/\tau$, with t the number of years since 2003 (Graf, 1977; Wu et al., 2012). Standard errors were computed with the heteroscedasticity- and autocorrelation-consistent (HAC) estimator of

Newey and West (1987), and uncertainty in τ with a moving-block bootstrap (Künsch, 1989). Measurement uncertainty, capacity estimates and budget closure are described in Appendix A.

4.2. Pathway decomposition

For each plateau station, Eqs. (16) and (17) were fitted by ordinary least squares. Temperature was centred on its station mean and precipitation normalized by its mean, so that c_T and $\partial \ln Q/\partial T$ are sensitivities per degree and c_P and ε_P^Q are elasticities. The total sensitivities on the left of Eq. (18) were estimated separately by regressing $\ln S$ on T' and $\ln P'$ alone; agreement between the total sensitivity and the sum of pathways checks the log-linear structure of Eqs. (16)–(17).

Uncertainty was estimated with a moving-block bootstrap (Künsch, 1989; Efron and Tibshirani, 1993). In each of 2000 replicates, years were resampled in blocks of 3 consecutive years, preserving serial correlation and the joint distribution of Q, S, T and P. All coefficients were re-estimated, and the product $b\,\partial \ln Q/\partial T$ was computed within each replicate so that its interval reflects uncertainty in both factors. An effect was considered detectable when its 95% percentile interval excluded zero. HAC p-values for c_T were adjusted for multiple testing with the false-discovery-rate procedure of Benjamini and Hochberg (1995). Stations whose single-variable rating explained less than 30% of the variance in $\ln S$ ($R^2 < 0.3$) were classed as poorly rated; for these, b is weakly constrained and the decomposition is not interpreted. The analysis was repeated with thresholds of $R^2 \geq 0.5$ and $R^2 \geq 0.7$.

Four further diagnostics were applied. (i) Collinearity among $\ln Q$, T' and $\ln P'$ was quantified with variance inflation factors. (ii) Statistical power for the direct temperature effect was expressed as the minimum detectable effect at 5% significance and 80% power, $\mathrm{MDE} = 2.8\,\mathrm{SE}(c_T)$. (iii) Lagged and cumulative effects of warming were tested by adding to Eq. (16) either the previous year's temperature anomaly or its 3-year running mean, and slow changes in availability by adding a linear time trend. (iv) Supply exhaustion was tested by adding the previous year's runoff to the rating, $\ln S_t = a + b \ln Q_t + b_{-1} \ln Q_{t-1}$: under exhaustion, a wet preceding year depletes available sediment and $b_{-1} < 0$, whereas under capacity control $b_{-1} \approx 0$. To test whether interannual exponents apply to long-term change, b was re-estimated from year-to-year differences, $\Delta \ln S$ on $\Delta \ln Q$, which emphasise high frequencies, and from non-overlapping 5-year means, which emphasise low frequencies.

4.3. Trends and rating change

Monotonic trends in T, Q, S and I_s were tested with the Mann–Kendall test (Mann, 1945; Kendall, 1975), and their magnitudes estimated with the Theil–Sen slope (Sen, 1968). The baseline rating in Eq. (20) was fitted to the first half of each record. Changes in rating shape were tested by fitting $\ln S = a + b_1 \ln Q + \delta_a L + \delta_b L \ln Q$, where $L = 1$ in the second half of the record and 0 otherwise, with HAC standard errors; $b_2 = b_1 + \delta_b$, and a change was considered significant when $p(\delta_b) < 0.05$. Spearman rank correlation was used to relate station-level anomalies and pathways to warming and to the thermal share of runoff variability, defined as $|\partial \ln Q/\partial T|\,\sigma_T / (|\partial \ln Q/\partial T|\,\sigma_T + |\varepsilon_P^Q|\,\sigma_{\ln P})$, where σ denotes the interannual standard deviation.

4.4. Plateau reach test

The only plateau pair in the dataset on a single main stem without major diversions is the upper Indus between Partab Bridge and Besham Qila. Equations (22) and (11), the latter without the time term, were fitted to this pair with Q at Besham Qila, to test whether outflow responds to inflow as predicted for a starved reach ($\lambda \approx 1$, $\beta_S \approx \bar{S}_{in}/\bar{S}_{out}$) or for a capacity-controlled reach ($\lambda \approx 0$, $\beta_S \approx 0$). Pairs on the Shule and Tarim rivers were not used because irrigation diversions remove 72–77% of the water between gauges. The difference between the transmission coefficients of the Jingjiang reach and the upper Indus, $\Delta\lambda$, was tested with a moving-block bootstrap (block length 3 yr, 2000 replicates) in which both fits were repeated on independently resampled years; its 95% percentile interval and a two-sided bootstrap p-value are reported.

4.5. Out-of-period test

The pathway decomposition, Eq. (18), is an algebraic property of linear projections fitted to the same data, so agreement between total and pathway elasticities tests internal consistency, not predictive skill. For an independent test, each record was split into two halves. The rating was fitted to the first half and used to predict (i) annual sediment flux in the second half from its runoff and (ii) the change in mean log sediment flux between halves, $\Delta \ln S_{\mathrm{pred}} = b_{\mathrm{train}}\, \Delta \ln Q$. Predictions were compared with observations by correlation and root-mean-square error. Departures of the observed change from the prediction measure changes in availability not captured by runoff.

5. Results

5.1. The dammed reach

Load at Shigu, the plateau outlet of the Yangtze, remained about 30 $\mathrm{Mt\,yr^{-1}}$ until 2019, whereas load at Pingshan, below the Jinsha cascade, fell from 139 $\mathrm{Mt\,yr^{-1}}$ in 2005–2012 to about 1.5 $\mathrm{Mt\,yr^{-1}}$ in 2014–2019 (Fig. 3a). Inflow to the Jingjiang reach fell from about 500 $\mathrm{Mt\,yr^{-1}}$ in the 1980s to 4 $\mathrm{Mt\,yr^{-1}}$ in 2022. The large reduction in sediment inflow, together with the sustained bed erosion reported below, indicates a sediment-starved reach.

Net exchange in the Zhicheng–Jianli reach fluctuated around zero before impoundment (mean -5.1 $\mathrm{Mt\,yr^{-1}}$, 1980–2002) and was positive in every year of 2004–2022 (mean 34.6 $\mathrm{Mt\,yr^{-1}}$), amounting to 678 Mt over 2003–2022.

In the structural model, Eq. (11), the inflow coefficient was $\lambda = 0.96 \pm 0.09$ for the whole reach, 1.03 ± 0.06 for the upper sub-reach and 0.88 ± 0.11 for the lower sub-reach (Table 3). Each is consistent with the starved-limit value of 1 and far from the capacity-limit value of 0. The log-linear supply elasticity of the whole reach was $\beta_S = 0.51 \pm 0.03$ for 2004–2022, within the range of the observed share of inflow in outflow (ratio of means 0.50; mean of annual ratios 0.41; median 0.47; annual range 0.17–0.71), and the partial discharge elasticity at fixed inflow was small ($\beta_Q = 0.15 \pm 0.14$). Before impoundment the share was 1.01 and $\beta_S = 0.96 \pm 0.04$. In the sub-reaches, β_S was 0.76 ± 0.07 (upper; ratio of means 0.70) and 0.63 ± 0.04 (lower; 0.68). With a linear time trend added to Eq. (22), the whole-reach β_S fell to 0.38 ± 0.05, indicating that the log-linear elasticity partly absorbs the decline of bed release. Bed erosion was positively correlated with inflow in the whole reach ($r = 0.65$) and the upper sub-reach ($r = 0.85$), because both are largest in wet years. The agreement between β_S and the inflow share is therefore approximate, and λ is the more robust test. Collinearity between $\ln Q$ and $\ln S_{in}$ was modest ($r = 0.36$). With cumulative forcing in place of calendar time, λ was unchanged (whole reach 0.97 ± 0.08, upper 1.03 ± 0.05, lower 0.89 ± 0.10) and the fits were practically identical (ΔAICc ≤ 0.9), with decay scales $1/k_e$ of 30, 14 and 47 mean-flow years and exponents m of 1.21, 2.09 and 1.13. The calendar-time approximation is adequate for these records.

Bed release declined at different rates along the reach (Fig. 3b). In the upper, sand–gravel sub-reach, erosion decayed with $\tau = 11.2$ yr (block bootstrap 9.2–14.3 yr) and depended strongly on discharge ($m = 2.4 \pm 0.4$). In the lower, sand-bed sub-reach, erosion was nearly constant (20.4 $\mathrm{Mt\,yr^{-1}}$ in 2004–2014 and 19.7 $\mathrm{Mt\,yr^{-1}}$ in 2015–2022; Mann–Kendall $p = 0.89$).

5.2. Plateau ratings

Air temperature increased at all 16 plateau stations, significantly at 13, at rates of 0.13–0.65 °C per decade (Fig. 4a; Table 4). The largest trends occurred in the Tarim tributaries and on the Yarlung Tsangpo, and the smallest, 0.13–0.20 °C per decade, in the upper Indus. Runoff increased at most stations but significantly only at Changmabao and Yugo; it decreased significantly at Dainyor.

Eleven stations had well-defined ratings ($R^2 \geq 0.3$). At all of them the rating exponent exceeded 1, with a median of 2.29 and a range of 1.42–3.30 (Fig. 4b). These values lie within the range expected under capacity control at the annual scale (Section 2.4) and well above the value of about 1 expected under supply control. The five poorly rated stations, Xiangda on the upper Lancang, Alar on the Tarim main stem, and Yugo, Partab

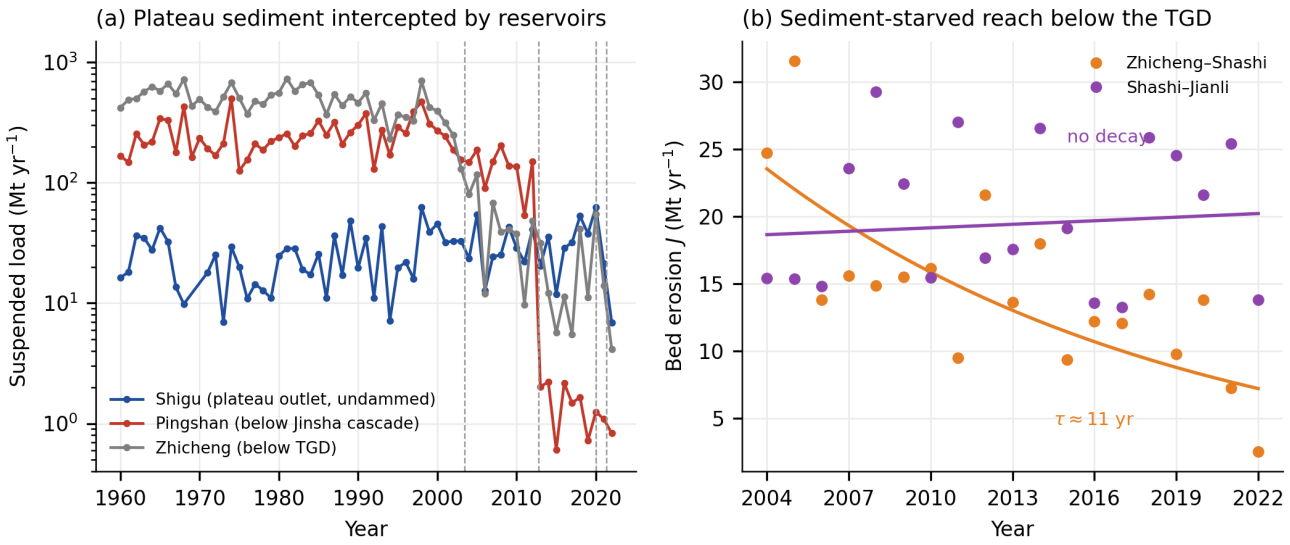


Fig. 3. (a) Suspended load at the plateau outlet of the Yangtze (Shigu), below the Jinsha cascade (Pingshan) and below the TGD (Zhicheng); dashed lines mark impoundments in 2003, 2012–2013, 2020 and 2021. (b) Net bed erosion of the two Jingjiang sub-reaches with fitted exponential trends.

Bridge and Besham Qila on the upper Indus, had exponents of 0.46–1.0 with wide intervals, which suggests that their sediment loads are controlled by factors other than runoff, such as landslides and glacial outburst floods, or by regulation and withdrawals. The median exponent was 2.29 for $R^2 \geq 0.3$ ($n = 11$), 2.33 for $R^2 \geq 0.5$ ($n = 8$) and 2.28 for $R^2 \geq 0.7$ ($n = 4$).

At 7 of the 11 well-rated stations, the previous year's runoff had no detectable effect on sediment flux at fixed current runoff. At the two Shule gauges the lagged coefficient was strongly negative ($b_{-1} = -0.78$ at Changmabao and -1.31 at Panjiazhuang; both $p < 0.001$), so that a wet year was followed by a year of lower sediment yield. At Daojieba and Dainyor the lagged coefficients were positive ($+0.88$ and $+1.13$). A negative antecedent dependence is consistent with sediment depletion but is not a unique diagnostic of it, because previous-year runoff can also act through channel adjustment, redistribution of stored sediment, correlated climate anomalies or regulation. Conversely, the absence of a negative lag does not prove abundant supply, since supply that recovers within a year would also give $b_{-1} \approx 0$.

The upper Indus pair gave $\lambda = 0.34 \pm 0.17$ and $\beta_S = 0.09 \pm 0.09$, well below the starved-limit values ($\lambda = 1$, $\beta_S \approx 0.70$). Both stations have poorly defined single-variable ratings ($R^2 < 0.3$), and the structural model explains only part of the variance in outflow ($R^2 = 0.15$; RMSE 19.7 against a mean outflow of 47.9 in the reported units). Within this limited fit, λ is separable from the discharge exponent ($\mathrm{corr}(\hat{\lambda}, \hat{m}) = -0.10$) but trades off with the reference bed release ($\mathrm{corr}(\hat{\lambda}, \ln \hat{B}_0) = -0.86$). A model with inflow only gave $\lambda = 0.38 \pm 0.12$ and a slightly lower AICc (246.1, against 247.6 for the full model and 249.2 for discharge only). Leave-one-year-out estimates ranged from 0.26 to 0.39, and block-bootstrap medians from 0.27 to 0.35 for block lengths of 1–5 yr, with upper 95% limits of 0.58–0.68. Separate fits to the two halves of the record gave 0.46 and 0.13. The estimate is therefore robustly below 1, but its lower bound is close to 0 for block lengths of 2 yr or more, and it varies between periods. The difference from the Jingjiang reach was $\Delta\lambda = 0.66$ (block bootstrap 95% interval 0.29–1.07; $p = 0.002$; Fig. 2c,d). The two reaches differ in hydrology, slope, grain size and morphology as well as in supply, so this difference is consistent with, but does not by itself isolate, a difference in transport regime.

Exponents estimated from year-to-year differences (median 2.34) and from 5-year means (median 2.37) were close to the annual estimate (2.29). Individual stations varied widely at low frequency, because only 4–10 five-year blocks were available, and refitting a power law to averaged data is affected by changes in flow variance, since $\overline{Q^b} \neq \overline{Q}^b$. A more direct check uses, for each well-rated station, the residual $R_i = \Delta \ln S_i - b_i \Delta \ln Q_i$

between record halves, where b_i is the full-record exponent. The residual measures the change in effective availability at fixed runoff, together with rating-shape changes and model error. Its median absolute value was 0.17, against a median absolute change in $\ln S$ of 0.36; six of the 11 stations had $|R_i| < 0.2$, and the largest residuals were at Dainyor (-0.78), Yanqi (-0.60), Panjiazhuang ($+0.46$), Lhasa ($+0.38$) and Daojieba ($+0.34$).

Ratings fitted to the first half of each record predicted annual sediment flux in the second half with a median correlation of 0.83 (0.70–1.00 at 10 of the 11 well-rated stations; 0.14 at Dainyor; Fig. 6c). The sign of the change in mean log sediment flux between halves was predicted correctly at 8 of 11 stations, and predicted and observed changes were correlated ($r = 0.81$; $r = 0.42$ without Dainyor). The largest departures from the predicted change were at Dainyor (-1.01), Yanqi (-0.54), Panjiazhuang ($+0.48$), Lhasa ($+0.38$) and Daojieba ($+0.35$), the same stations that stand out in the anomaly and rating-change diagnostics.

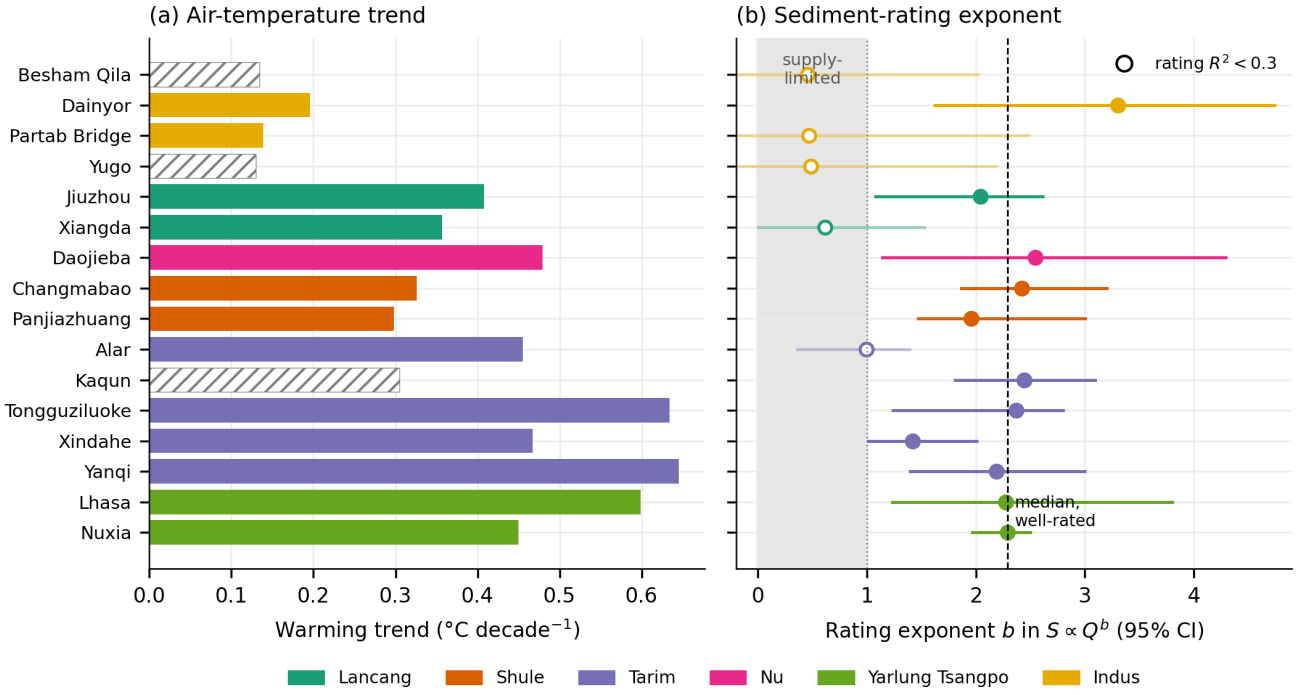


Fig. 4. (a) Theil–Sen trend in basin-mean air temperature; hatched bars are not significant ($p \geq 0.05$). (b) Rating exponent b with 95% block-bootstrap intervals. Open symbols: poorly rated stations ($R^2 < 0.3$). The grey band marks supply-limited ratings ($b < 1$); the dashed line is the median of the well-rated stations.

5.3. *Precipitation*

The total precipitation elasticity of sediment flux was detectable at six well-rated stations and ranged from 1.4 at Changmabao to 4.3 at Lhasa (Fig. 5a; Table 2). These values exceed the corresponding runoff elasticities, 0.5–1.8, by a factor of 2–3, as expected from Eq. (18) for $b \approx 2.3$. At Lhasa, Nuxia, Daojieba and Jiuzhou, $b\,\varepsilon_P^Q$ was 4.0, 3.2, 2.8 and 3.0, against total elasticities of 4.3, 3.0, 3.0 and 3.8, and the direct term c_P was indistinguishable from zero at each of these stations. In the monsoon-fed basins, the higher sediment flux of wet years is accounted for almost entirely by their higher runoff.

5.4. *Temperature*

The direct effect of temperature at fixed runoff, c_T, was not detectable at 9 of the 11 well-rated stations by the bootstrap criterion (Fig. 5b). The two exceptions, a positive effect at Lhasa ($+0.16$ °C^{-1}; 95% interval 0.02–0.71) and a negative effect at Dainyor (-0.47 °C^{-1}), were not significant after false-discovery-rate adjustment. The runoff pathway of temperature was positive and detectable at the glacier-fed Changmabao on the Shule

River ($b\,\partial \ln Q/\partial T = 0.37$ °C^{-1}; 0.15–0.61) and at Xindahe in the Tarim basin (0.17 °C^{-1}; 0.03–0.38), the two well-rated stations at which runoff itself increases detectably with temperature (0.15 and 0.12 °C^{-1}). Across stations, the thermal share of runoff variability was correlated with the runoff pathway of temperature (Spearman $\rho = 0.85$, $p = 0.001$). The total temperature sensitivity of sediment flux was detectable only at Changmabao (+0.25 °C^{-1}; 0.004–0.45), where it was accounted for by the runoff pathway.

Variance inflation factors for T' were 1.00–1.52, and the correlation between T' and $\ln Q$ exceeded 0.5 only at Changmabao and Nuxia, so collinearity does not explain the absence of a detectable direct effect. Power is limited: the minimum detectable direct effect had a median of 0.26 °C^{-1} (0.11–0.38 at the well-rated stations other than Dainyor), so direct effects smaller than about 25% per degree cannot be excluded at most stations. The previous year's temperature was significant at two stations and the 3-year mean temperature at five; four of the five 3-year coefficients were negative (Jiuzhou, Yanqi, Lhasa, Dainyor) and one positive (Daojieba). Linear trends in availability at fixed runoff, temperature and precipitation were negative at five well-rated stations and positive at one (Daojieba). At the annual scale, the detectable temperature-associated changes in sediment export are therefore mediated by runoff, and neither interannual, lagged nor multi-year warming is associated with a detectable increase in availability at fixed flow.

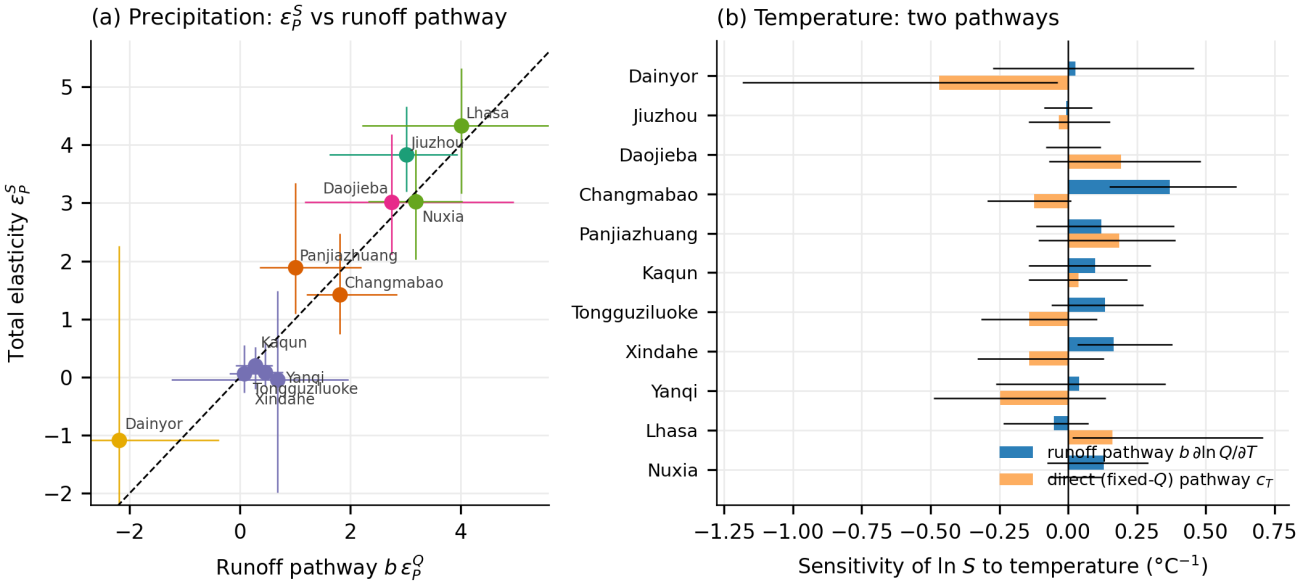


Fig. 5. (a) Total precipitation elasticity of sediment flux versus the runoff pathway $b\,\varepsilon_P^Q$ at the well-rated stations, with 95% block-bootstrap intervals; the dashed line is 1:1. (b) Temperature sensitivity of $\ln S$ decomposed into the runoff pathway ($b\,\partial \ln Q/\partial T$) and the direct, fixed-discharge pathway (c_T), with 95% intervals.

Table 2. Pathway results at the well-rated stations ($R^2 \geq 0.3$). b: rating exponent; ε_P^Q, ε_P^S: precipitation elasticities of runoff and sediment flux; $\partial_T Q = \partial \ln Q/\partial T$ and c_T: temperature sensitivity of runoff and direct temperature effect on sediment flux at fixed Q (°C^{-1}). Bold: 95% block-bootstrap interval excludes zero. $b_1 \to b_2$: rating exponent in the first and second halves of the record; * change significant at $p < 0.05$.

River	Station	Period	b	ε_P^Q	ε_P^S	$\partial_T Q$	c_T	$b_1 \to b_2$
Lancang	Jiuzhou	1979–2008	2.04	**1.48**	**3.83**	−0.003	−0.03	2.41→2.59
Shule	Changmabao	1960–2012	2.42	**0.75**	**1.42**	**0.152**	−0.12	2.85→1.68*
	Panjiazhuang	1960–2012	1.96	**0.51**	**1.89**	0.061	0.19	3.48→1.92*
Tarim	Kaqun	1980–2010	2.44	0.11	0.21	0.040	0.04	2.03→3.13
	Tongguziluoke	1980–2010	2.37	0.04	0.07	0.056	−0.14	1.70→3.12*
	Xindahe	1980–2010	1.42	**0.32**	0.09	**0.117**	−0.14	1.28→1.00
	Yanqi	1980–2010	2.19	0.31	−0.04	0.018	−0.25	1.22→2.57*
Nu	Daojieba	1979–2010	2.55	**1.08**	**3.01**	0.000	0.19	2.42→2.47
Yarlung Tsangpo	Lhasa	1980–2010	2.27	**1.77**	**4.33**	−0.023	**0.16**	2.25→2.28
	Nuxia	1978–2000	2.29	**1.39**	**3.03**	0.056	0.00	2.20→2.29
Indus	Dainyor	1966–2012	3.30	−0.66	−1.09	0.008	**−0.47**	2.35→0.85

5.5. Availability change

The discharge-conditioned anomaly I_s increased significantly at 3 of the 18 stations (Panjiazhuang, Daojieba and Shigu) and decreased at 6 (Changmabao, Tongguziluoke, Yanqi, Alar, Dainyor and Partab Bridge) (Fig. 6a). Its mean in the second half of each record was not correlated with the magnitude of warming across stations (Spearman $\rho = 0.08$, $p = 0.78$) or with the change in precipitation ($\rho = 0.19$, $p = 0.48$).

The rating exponent changed significantly at four stations, in opposite directions (Fig. 6b). On the glacier-fed Shule River it decreased at Changmabao ($2.85 \rightarrow 1.68$) and Panjiazhuang ($3.48 \rightarrow 1.92$), so that high-flow years carried proportionally less sediment than before. In the Tarim basin it increased at Tongguziluoke ($1.70 \rightarrow 3.12$) and Yanqi ($1.22 \rightarrow 2.57$). At Shigu the rating was steep ($b = 2.97$) and I_s increased slightly but significantly through 2022.

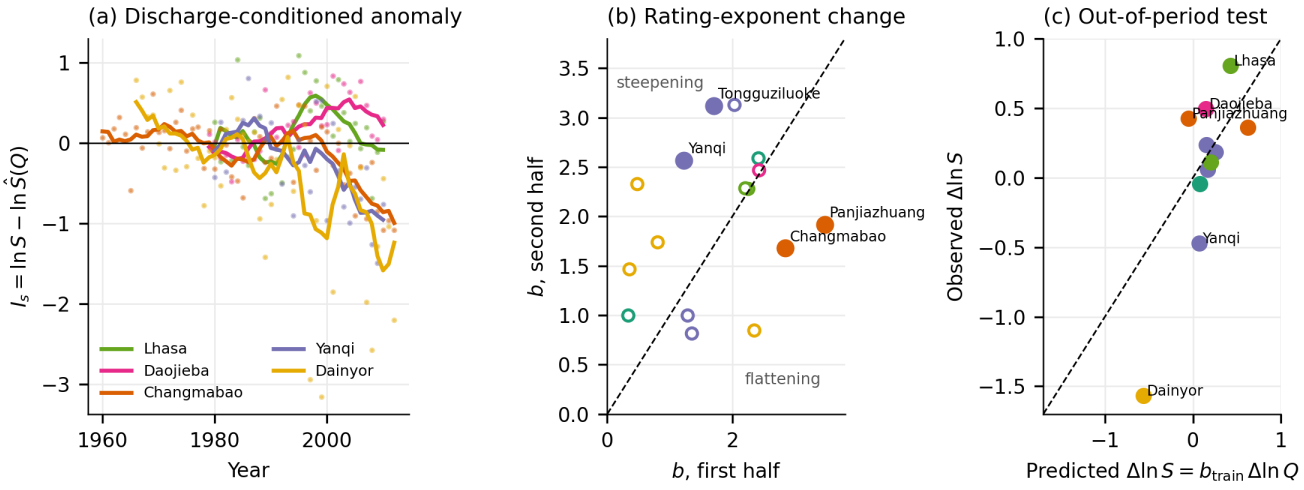


Fig. 6. (a) Discharge-conditioned sediment anomaly I_s at five representative stations; dots are annual values and lines 7-yr running means. (b) Rating exponent in the first versus the second half of each record; filled symbols denote significant changes ($p < 0.05$). (c) Out-of-period test at the well-rated stations: change in mean log sediment flux between record halves predicted from the first-half rating and the observed change in runoff, versus the observed change; the dashed line is 1:1, and labelled stations depart by more than 0.3.

5.6. Comparison of the two systems

Table 3 summarises the results for both systems. On the plateau, ratings are steep, the precipitation response passes through runoff, no direct climate effect on availability is detectable, and the only gauged plateau reach has $\lambda = 0.34$. Below the dams, λ is indistinguishable from 1 and the discharge elasticity from 0. The two systems lie towards opposite limits of Eq. (4), with the plateau closer to, but not at, the sediment-rich limit.

Table 3. Signatures of the two limits of Eq. (4). λ: inflow coefficient of the structural model, Eq. (11) (starved limit 1, sediment-rich limit 0); β_Q, β_S: log-linear elasticities of outflow with respect to discharge and inflow (Eq. 22); share: ratio of mean inflow to mean outflow. Uncertainties are standard errors.

System	λ	β_S	β_Q	Share
Plateau, 11 well-rated stations	–	–	$b = 2.29$ (median)	–
Upper Indus, Partab Br. $\rightarrow$ Besham Q.	0.34 ± 0.17	0.09 ± 0.09	0.76 ± 0.54	0.70
Jingjiang, 1980–2002	–	0.96 ± 0.04	-0.26 ± 0.20	1.01
Jingjiang, 2004–2022	0.96 ± 0.09	0.51 ± 0.03	0.15 ± 0.14	0.50
Zhicheng–Shashi	1.03 ± 0.06	0.76 ± 0.07	–	0.70
Shashi–Jianli	0.88 ± 0.11	0.63 ± 0.04	–	0.68

6. Discussion

6.1. Capacity control

At the annual scale, the climate sensitivity of plateau sediment export appears to be governed mainly by the climate sensitivity of runoff and by the steepness of the sediment rating. The evidence is as follows: ratings are steep at all well-rated stations; precipitation acts almost entirely through runoff; no direct, lagged or multi-year warming is associated with increased availability; most stations show no negative dependence on antecedent runoff; and the one gauged plateau reach passes inflow variations downstream only partially. None of these observations alone shows that the rivers are capacity-controlled, and a steep rating could arise from supply that increases nonlinearly with discharge. Their combination is consistent with capacity-controlled transport and is not easily explained if sediment availability sets the flux.

If this interpretation is correct, a given proportional change in runoff produces a two- to threefold larger proportional change in sediment flux. Applying Eq. (19) to the regional runoff trend reported by Li et al. (2021), an increase of 5% per decade with $b = 2.29$ gives $1.05^{2.29} - 1 \approx 0.118$, or about 12% per decade, the rate those authors observed. Runoff amplification is quantitatively sufficient to explain a trend of this magnitude. This is a consistency of magnitude, not an attribution of the historical increase; it does not show that availability has not changed, and the out-of-period test shows that it has changed at several stations, in both directions. The calculation also assumes that the interannual exponent applies to decadal change, which is supported by the similarity of high- and low-frequency exponents but cannot be tested rigorously with records of a few decades.

Transport theory predicts steep ratings under capacity control but not the observed value of b. Hydraulic geometry and the Engelund–Hansen formula give an instantaneous capacity exponent of about 1.5 (Eq. 14), and Eq. (15) attributes the remaining 0.75 to more peaked hydrographs in wet years. This contribution is plausible for monsoon- and melt-driven rivers but was not quantified here and remains a hypothesis to be tested with daily discharge records.

6.2. Fate of released sediment

If plateau rivers are largely capacity-controlled, the balance predicts that sediment mobilised by glacier retreat, permafrost thaw or rainstorms is exported only to the extent that flow can carry it, the remainder being deposited in channels, alluvial fans and floodplains. This is consistent with the estimate of Li et al. (2024b) that about 30% of the sediment mobilised in plateau headwaters is deposited within river channels, and with the absence of a fixed-flow climate signal in the station records. Channel storage may therefore buffer climate-induced increases in sediment availability.

It follows from the balance, though it is not shown by our data, that stored sediment could be released by future high flows, so that part of the export response to warming would be deferred and tied to large floods. Testing this requires direct estimates of the change in channel storage, dM/dt, and of the flows needed to remobilise it. If correct, projections based on temperature alone would overestimate near-term export and underestimate episodic releases, and on dammed rivers such as the Yangtze and the Mekong the deferred supply would reach reservoirs rather than the sediment-starved channels downstream.

6.3. Role of the downstream test

The Jingjiang reach is the most direct test of the balance available to us, because its supply was reduced abruptly and independently of climate. λ is indistinguishable from 1 in the whole reach and in both sub-reaches, as mass conservation with separable bed release requires, whereas the plateau reach gives $\lambda = 0.34$. The log-linear elasticity β_S lies within the observed range of the inflow share, although its value depends on how the share is summarised and on whether the decline of bed release is controlled for. Because the same methods detect supply dependence where it is known to exist, we interpret the absence of a direct climate effect on the plateau as consistent with capacity control rather than solely as a lack of power. The two reaches, however, differ in hydrology, slope, grain size, morphology and sediment residence time as well as in supply. The plateau estimate is also weakly constrained: it rests on one reach whose stations have poor single-variable ratings, its structural

fit explains little of the variance, and its value differs between the two halves of the record. The difference in λ is consistent with a difference in transport regime but does not directly establish capacity control on the plateau; that would require an independent estimate of S_c from measured hydraulics and bed material, or a comparison of λ within one reach before and after a change in supply. The minimum detectable effects also show that direct effects of up to about 25% per degree could have escaped detection at most plateau stations.

In the upper sub-reach, which has a sand–gravel bed, the erosion rate decreased by more than 40% within a decade, consistent with armouring and with the downstream migration of scour described by Lai et al. (2017). The lower sand-bed sub-reach shows no decay after two decades. The decline in coarse-sediment recovery documented by Zhang et al. (2017) and Yang et al. (2022) is consistent with the first behaviour, and the persistence of erosion in the meandering Lower Jingjiang (Xia et al., 2016; Lyu et al., 2019) with the second.

6.4. Previous work

The components of the analysis are standard: the competition between supply and capacity (Lane, 1955; Church, 2006), sediment continuity (Exner, 1925; Paola and Voller, 2005), the rating relation (Asselman, 2000; Syvitski et al., 2000) and the elasticity concept (Schaake, 1990; Sankarasubramanian et al., 2001). They are combined here in two ways. First, the sensitivity of sediment flux to runoff and to supply reverses within one river system, from the plateau to the reaches below its dams, and both behaviours correspond to limits of one supply–capacity–storage balance whose starved limit was tested quantitatively. Second, the climate sensitivity of plateau sediment flux is partitioned into runoff and availability pathways. Li et al. (2021) estimated total climate elasticities and attributed the regional increase to sediment release; we find that runoff amplification is quantitatively sufficient to explain an increase of the reported magnitude, and that the glacier dependence of their temperature sensitivities is consistent with warming acting through runoff, which is more temperature-sensitive in glacier-fed basins. Li et al. (2024a) identified glacier cover as the main spatial control on sediment yield; our analysis concerns the temporal sensitivity of flux, which can be dominated by runoff even where glaciers set the long-term yield. Zhang et al. (2023) documented seasonal regime shifts in four cryosphere-fed basins; at the annual scale we find behaviour consistent with capacity control at most stations and evidence of supply limitation on the glacier-fed Shule. The conclusion of Zhang et al. (2020) that the sediment increase in the Tuotuo River reflects transport capacity is consistent with ours.

6.5. Basin-specific changes

Three basins depart from the regional pattern. On the glacier-fed Shule River the rating flattened at both gauges, availability at fixed flow declined at Changmabao, and at both gauges wet years were followed by years of lower sediment yield. These observations are consistent with progressive exhaustion of a finite sediment supply, a stage of paraglacial adjustment (Ballantyne, 2002), possibly combined with dilution of high flows by relatively clear meltwater and with the effects of reservoirs and diversions. On the Shule, warming appears to increase sediment export through runoff while availability declines. In the Tarim basin, the steepening of the ratings at Tongguziluoke and Yanqi suggests that high flows increasingly access stored or newly exposed sediment. At Daojieba on the Nu River, availability at fixed flow increased and wet years were followed by higher yields, consistent with delayed delivery of sediment mobilised by earlier events.

6.6. Identifiability of transport regimes

None of the diagnostics used here identifies the transport regime on its own. A steep rating ($b > 1$) shows that sediment flux is highly sensitive to runoff but does not establish capacity control, because supply can increase nonlinearly with discharge. A negative dependence on antecedent runoff ($b_{-1} < 0$) is consistent with depletion but has other possible causes, and its absence does not prove abundant supply. A transmission coefficient below one ($\lambda < 1$) shows that inflow variations are not fully passed downstream but can reflect storage, selective deposition or travel times as well as capacity limitation. The absence of a detectable direct climate effect ($c_T \approx 0$) is limited by statistical power. The interpretation of plateau rivers as predominantly capacity-controlled rests on the agreement among these diagnostics and on the contrast with a reach where

supply limitation is known. It applies to annual records; at the event scale, hysteresis and exhaustion within floods may indicate supply limitation even in rivers that are capacity-controlled on average (Asselman, 2000). Estimates of transport capacity from measured hydraulic geometry and bed grain size would allow Π_s to be evaluated directly.

6.7. Limitations

Reach averaging conceals the downstream propagation of scour that the sub-reach budgets reveal in the Jingjiang reach; a finer division would probably show further structure. Because of the annual time scale, rainstorms, glacial lake outburst floods and landslides, which deliver a large share of sediment in mountain rivers, enter only through their effect on annual runoff and as unexplained variance, which is largest at the poorly rated stations. The use of total suspended load as a single effective fraction, Eqs. (6)–(7), is the main reason the downstream budgets account for only about one-third of surveyed channel erosion (Appendix A); bedload, bank erosion and sand mining are not resolved. The separability assumption holds approximately in the whole reach and the lower sub-reach, but less well in the upper sub-reach, where bed erosion is strongly correlated with inflow. Decadal inferences rest on records of a few decades and on only two record halves per station. Stationarity of basin conditions is doubtful at the lowland Tarim and Shule stations, where diversions are large, and at stations affected by regulation. Station coordinates in Fig. 1 are approximate, and basin attributes such as glacier and permafrost cover (RGI Consortium, 2017) were not included; relating the direct pathway and the rating changes to these attributes would test whether the departures from the regional pattern are linked to cryospheric change.

7. Conclusions

1. In the Jingjiang reach below the Three Gorges Dam, outflow followed inflow with $\lambda = 0.96 \pm 0.09$ (0.97 ± 0.08 with cumulative forcing), as mass conservation requires in a sediment-starved reach; the partial discharge elasticity at fixed inflow was 0.15 ± 0.14. Bed erosion totalled 678 Mt over 2003–2022 and decayed with $\tau = 11.2$ yr (9.2–14.3 yr) in the armouring upper sub-reach, with no detectable decay in the sandy lower sub-reach.

2. In the upper Indus, the only gauged plateau reach, $\lambda = 0.34 \pm 0.17$ ($\Delta\lambda = 0.66$, 95% interval 0.29–1.07). The estimate is robustly below 1 but weakly constrained.

3. Plateau sediment ratings have a median exponent of 2.29 (1.42–3.30; $n = 11$), insensitive to the station-selection threshold. Seven of the 11 well-rated stations show no detectable dependence on antecedent runoff; the glacier-fed Shule River shows a negative dependence consistent with supply depletion.

4. Precipitation elasticities of sediment flux (1.4–4.3) exceed those of runoff by a factor of 2–3 and are reproduced by the runoff pathway. No direct temperature effect at fixed runoff is detectable after multiple-testing adjustment (median minimum detectable effect 0.26 $^{\circ}\mathrm{C}^{-1}$), and no lagged or multi-year warming is associated with increased availability.

5. With $b = 2.29$, a runoff increase of 5% per decade gives $(1.05)^{2.29} - 1 \approx 11.8\%$ per decade in sediment flux. The observations support runoff amplification as a sufficient explanation for the reported regional increase, while leaving slower changes in sediment supply unresolved.

CRediT authorship contribution statement

Weixiang Zhao: Data curation, Formal analysis, Investigation, Software, Visualization, Writing – original draft. **Huilin Wang:** Conceptualization, Methodology, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Yangtze station data were compiled from the *China River Sediment Bulletin*. Compiled series, derived tables and analysis code are provided as supplementary material.

Acknowledgements

We thank the authors of the original datasets for making their data available, and Shawn Li at Columbia University for useful discussions.

Appendix A. Downstream sediment budgets

Sub-reach budgets. Sub-reach exchange was computed as $J_{ZS} = S_{SS} + S_{XJK} + S_{SDG} + S_{MTS} - S_{ZC}$ and $J_{SJ} = S_{JL} + S_{KJG} + S_{GJP} - S_{SS}$, where subscripts denote Zhicheng (ZC), Shashi (SS), Jianli (JL), Xinjiangkou (XJK), Shadaoguan (SDG), Mituosi (MTS), Kangjiagang (KJG) and Guanjiapu (GJP).

Measurement uncertainty. Budgets computed as small differences between large fluxes are sensitive to measurement error (Kondolf and Matthews, 1991). Assuming independent relative errors ε in each annual load, the standard error of J is $\sigma_J = \varepsilon(\sum_i S_i^2)^{1/2}$. Calibrating ε so that pre-dam normalized residuals have unit root-mean-square gives $\varepsilon = 5.3\%$, an upper bound because pre-dam residuals include real deposition and erosion. Post-dam uncertainties for the whole reach are 1.1–9.7 $\mathrm{Mt\,yr^{-1}}$, well below the mean erosion of 34.6 $\mathrm{Mt\,yr^{-1}}$.

Decay of erosion. For the whole reach, the rate law gave $\tau = 28$ yr (block bootstrap 22–42 yr; 37.5 yr excluding the 2022 drought). This value is a composite of the two sub-reaches and does not correspond to a single physical time scale. Residuals of the whole-reach fit showed no serial correlation (Durbin–Watson 1.99; Durbin and Watson, 1950; Ljung–Box $p > 0.37$; Ljung and Box, 1978).

Transport capacity. A historical rating benchmark fitted to pre-dam outflow implied residual capacities of about 360 $\mathrm{Mt\,yr^{-1}}$, an order of magnitude above an Engelund–Hansen (Engelund and Hansen, 1967) scoping estimate of 16–50 $\mathrm{Mt\,yr^{-1}}$ for the sand-bed lower sub-reach, obtained by Monte Carlo sampling of $h = 8$–12 m, $B = 1000$–1500 m, $S_f = 3$–6×10^{-5} and $d_{50} = 0.15$–0.25 mm at the post-dam mean discharge (bankfull depths from Xia et al., 2016). The benchmark probably overestimates capacity because it includes supply-controlled wash load (Einstein, 1950). Observed erosion in the lower sub-reach (20 $\mathrm{Mt\,yr^{-1}}$) lies within the scoping range.

Budget closure. Cumulative suspended-load erosion over 2003–2020 (629 Mt) is 32–39% of the surveyed bankfull-channel erosion of the longer Zhicheng–Chenglingji reach, 12.29×10^8 $\mathrm{m^3}$ converted with a dry bulk density of 1.3–1.6 t $\mathrm{m^{-3}}$ (Ministry of Water Resources of China, 2020). The difference is attributed to the longer surveyed reach and to unmeasured exchange through bedload, bank erosion and sand mining.

Appendix B. Station summary

References

Asselman, N.E.M., 2000. Fitting and interpretation of sediment rating curves. J. Hydrol. 234, 228–248.
Ballantyne, C.K., 2002. Paraglacial geomorphology. Quat. Sci. Rev. 21, 1935–2017.
Baron, R.M., Kenny, D.A., 1986. The moderator–mediator variable distinction in social psychological research: Conceptual, strategic, and statistical considerations. J. Pers. Soc. Psychol. 51, 1173–1182.
Benjamini, Y., Hochberg, Y., 1995. Controlling the false discovery rate: A practical and powerful approach to multiple testing. J. R. Stat. Soc. B 57, 289–300.
Church, M., 2006. Bed material transport and the morphology of alluvial river channels. Annu. Rev. Earth Planet. Sci. 34, 325–354.
Dai, Z., Lu, X.X., 2014. Sediment load change in the Yangtze River (Changjiang): A review. Geomorphology 215, 60–73.
Durbin, J., Watson, G.S., 1950. Testing for serial correlation in least squares regression. I. Biometrika 37, 409–428.

Table 4. All stations. R^2: variance explained by the single-variable rating; b with 95% block-bootstrap interval; T trend in °C per decade († not significant); Mann–Kendall τ_K for Q, S and I_s (* $p < 0.05$).

River	Station	Period	n	R^2	b	T trend	$\tau_K(Q)$	$\tau_K(S)$	$\tau_K(I_s)$
Lancang	Xiangda	1959–2010	52	0.04	0.62 [0.00, 1.52]	0.36	+0.13	+0.06	+0.05
	Jiuzhou	1979–2008	30	0.74	2.04 [1.08, 2.62]	0.41	+0.12	−0.03	−0.24
Shule	Changmabao	1960–2012	53	0.67	2.42 [1.87, 3.20]	0.33	+0.43*	+0.22*	−0.36*
	Panjiazhuang	1960–2012	53	0.52	1.96 [1.47, 3.01]	0.30	+0.05	+0.23*	+0.19*
Tarim	Kaqun	1980–2010	31	0.72	2.44 [1.81, 3.10]	0.30†	+0.15	+0.06	−0.08
	Tongguziluoke	1980–2010	31	0.62	2.37 [1.24, 2.80]	0.63	+0.12	−0.07	−0.31*
	Xindahe	1980–2010	31	0.33	1.42 [1.01, 2.01]	0.47	+0.16	−0.10	−0.21
	Yanqi	1980–2010	29	0.44	2.19 [1.39, 3.00]	0.65	+0.06	−0.29*	−0.41*
	Alar	1980–2010	31	0.26	1.00 [0.36, 1.39]	0.46	+0.19	−0.31*	−0.49*
Nu	Daojieba	1979–2010	32	0.56	2.55 [1.14, 4.29]	0.48	+0.05	+0.31*	+0.41*
Yarlung Tsangpo	Lhasa	1980–2010	31	0.73	2.27 [1.23, 3.80]	0.60	+0.17	+0.14	+0.08
	Nuxia	1978–2000	23	0.97	2.29 [1.97, 2.50]	0.45	+0.16	+0.09	−0.27
Indus	Yugo	1974–2013	40	0.06	0.49 [-1.07, 2.18]	0.13†	+0.25*	−0.13	−0.14
	Dainyor	1966–2012	47	0.33	3.30 [1.62, 4.74]	0.20	−0.34*	−0.63*	−0.33*
	Partab Bridge	1964–2008	45	0.00	0.47 [-1.99, 2.49]	0.14	+0.17	−0.23*	−0.28*
	Besham Qila	1969–2008	40	0.06	0.46 [-1.18, 2.02]	0.13†	+0.09	−0.17	−0.18
Jinsha	Batang	1964–2022	47	0.63	2.45	–	–	–	+0.10
	Shigu	1958–2022	63	0.81	2.97	–	–	–	+0.34*

Efron, B., Tibshirani, R.J., 1993. An Introduction to the Bootstrap. Chapman & Hall, New York.

Einstein, H.A., 1950. The bed-load function for sediment transportation in open channel flows. Technical Bulletin 1026, U.S. Department of Agriculture, Washington, DC.

Engelund, F., Hansen, E., 1967. A Monograph on Sediment Transport in Alluvial Streams. Teknisk Forlag, Copenhagen.

Exner, F.M., 1925. Über die Wechselwirkung zwischen Wasser und Geschiebe in Flüssen. Sitzungsber. Akad. Wiss. Wien, Math.-Naturwiss. Kl., Abt. IIa 134, 165–204.

Graf, W.L., 1977. The rate law in fluvial geomorphology. Am. J. Sci. 277, 178–191.

Guo, L., Su, N., Townend, I., Wang, Z.B., Zhu, C., Wang, X., Zhang, Y., He, Q., 2019. From the headwater to the delta: A synthesis of the basin-scale sediment load regime in the Changjiang River. Earth-Sci. Rev. 197, 102900.

Hallet, B., Hunter, L., Bogen, J., 1996. Rates of erosion and sediment evacuation by glaciers: A review of field data and their implications. Global Planet. Change 12, 213–235.

Hirano, M., 1971. River bed degradation with armoring. Proc. Jpn. Soc. Civ. Eng. 195, 55–65.

Horowitz, A.J., 2003. An evaluation of sediment rating curves for estimating suspended sediment concentrations for subsequent flux calculations. Hydrol. Process. 17, 3387–3409.

Immerzeel, W.W., van Beek, L.P.H., Bierkens, M.F.P., 2010. Climate change will affect the Asian water towers. Science 328, 1382–1385.

Kendall, M.G., 1975. Rank Correlation Methods. Charles Griffin, London.

Kondolf, G.M., Matthews, W.V.G., 1991. Unmeasured residuals in sediment budgets: A cautionary note. Water Resour. Res. 27, 2483–2486.

Kondolf, G.M., Gao, Y., Annandale, G.W., et al., 2014. Sustainable sediment management in reservoirs and regulated rivers: Experiences from five continents. Earth's Future 2, 256–280.

Kuang, X., Jiao, J.J., 2016. Review on climate change on the Tibetan Plateau during the last half century. J. Geophys. Res. Atmos. 121, 3979–4007.

Künsch, H.R., 1989. The jackknife and the bootstrap for general stationary observations. Ann. Stat. 17, 1217–1241.

Lai, X., Yin, D., Finlayson, B.L., et al., 2017. Will river erosion below the Three Gorges Dam stop in the middle Yangtze? J. Hydrol. 554, 24–31.

Lane, E.W., 1955. The importance of fluvial morphology in hydraulic engineering. Proc. Am. Soc. Civ. Eng. 81 (745), 1–17.

Leopold, L.B., Maddock, T., 1953. The hydraulic geometry of stream channels and some physiographic implications. U.S. Geological Survey Professional Paper 252.

Li, D., Zhang, T., Walling, D.E., Lane, S., Bookhagen, B., Tian, S., Overeem, I., Syvitski, J., Kettner, A.J., Park, E., Koppes, M., Schmitt, R.J.P., Sun, W., Ni, J., Ehlers, T.A., 2024. The competing controls of glaciers, precipitation, and vegetation on high-mountain fluvial sediment yields. Sci. Adv. 10, eads6196.

Li, J., Wang, G., Song, C., Sun, S., Ma, J., Wang, Y., Guo, L., Li, D., 2024. Recent intensified erosion and massive sediment deposition in Tibetan Plateau rivers. Nat. Commun. 15, 722.

Li, D., Lu, X., Walling, D.E., et al., 2021. Exceptional increases in fluvial sediment fluxes in a warmer and wetter High Mountain Asia. Science 374, 599–603.

Ljung, G.M., Box, G.E.P., 1978. On a measure of lack of fit in time series models. Biometrika 65, 297–303.

Lyu, Y., Zheng, S., Tan, G., Shu, C., Han, Q., 2019. Morphodynamic adjustments in the Yichang–Chenglingji reach of the Middle Yangtze River since the operation of the Three Gorges Project. Catena 172, 274–284.

Mann, H.B., 1945. Nonparametric tests against trend. Econometrica 13, 245–259.

Milliman, J.D., Syvitski, J.P.M., 1992. Geomorphic/tectonic control of sediment discharge to the ocean: The importance of small mountainous rivers. J. Geol. 100, 525–544.

Ministry of Water Resources of China, 2020. China River Sediment Bulletin 2020. China Water & Power Press, Beijing (in Chinese).

Ministry of Water Resources of China, 2000–2022. China River Sediment Bulletin (annual issues). China Water & Power Press, Beijing (in Chinese).

Newey, W.K., West, K.D., 1987. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. Econometrica 55, 703–708.

Paola, C., Voller, V.R., 2005. A generalized Exner equation for sediment mass balance. J. Geophys. Res. Earth Surf. 110, F04014.

Parker, G., Klingeman, P.C., 1982. On why gravel bed streams are paved. Water Resour. Res. 18, 1409–1423.

RGI Consortium, 2017. Randolph Glacier Inventory – A Dataset of Global Glacier Outlines, Version 6.0. GLIMS Technical Report, Boulder, Colorado.

Rubin, D.M., Topping, D.J., 2001. Quantifying the relative importance of flow regulation and grain size regulation of suspended sediment transport (α) and tracking changes in grain size of bed sediment (β). Water Resour. Res. 37, 133–146.

Sankarasubramanian, A., Vogel, R.M., Limbrunner, J.F., 2001. Climate elasticity of streamflow in the United States. Water Resour. Res. 37, 1771–1781.

Schaake, J.C., 1990. From climate to flow. In: Waggoner, P.E. (Ed.), Climate Change and U.S. Water Resources. John Wiley, New York, pp. 177–206.

Sen, P.K., 1968. Estimates of the regression coefficient based on Kendall's tau. J. Am. Stat. Assoc. 63, 1379–1389.

Syvitski, J.P.M., Milliman, J.D., 2007. Geology, geography, and humans battle for dominance over the delivery of fluvial sediment to the coastal ocean. J. Geol. 115, 1–19.

Syvitski, J.P., Morehead, M.D., Bahr, D.B., Mulder, T., 2000. Estimating fluvial sediment transport: The rating parameters. Water Resour. Res. 36, 2747–2760.

Topping, D.J., Rubin, D.M., Vierra, L.E., 2000. Colorado River sediment transport: 1. Natural sediment supply limitation and the influence of Glen Canyon Dam. Water Resour. Res. 36, 515–542.

Walling, D.E., 1977. Assessing the accuracy of suspended sediment rating curves for a small basin. Water Resour. Res. 13, 531–538.

Walling, D.E., 2006. Human impact on land–ocean sediment transfer by the world's rivers. Geomorphology 79, 192–216.

Warrick, J.A., 2015. Trend analyses with river sediment rating curves. Hydrol. Process. 29, 1083–1099.

Wu, B., Zheng, S., Thorne, C.R., 2012. A general framework for using the rate law to simulate morphological response to disturbance in the fluvial system. Prog. Phys. Geogr. 36, 575–597.

Xia, J., Deng, S., Lu, J., Xu, Q., Zong, Q., Tan, G., 2016. Dynamic channel adjustments in the Jingjiang Reach of the Middle Yangtze River. Sci. Rep. 6, 22802.

Yang, S.L., Milliman, J.D., Li, P., Xu, K., 2011. 50,000 dams later: Erosion of the Yangtze River and its delta. Global Planet. Change 75, 14–20.

Yang, S.L., Milliman, J.D., Xu, K.H., Deng, B., Zhang, X.Y., Luo, X.X., 2014. Downstream sedimentary and geomorphic impacts of the Three Gorges Dam on the Yangtze River. Earth-Sci. Rev. 138, 469–486.

Yang, S.L., Xu, K.H., Milliman, J.D., Yang, H.F., Wu, C.S., 2015. Decline of Yangtze River water and sediment discharge: Impact from natural and anthropogenic changes. Sci. Rep. 5, 12581.

Yang, Y., Zheng, J., Zhu, L., Zhang, H., Wang, J., 2022. Influence of the Three Gorges Dam on the transport and sorting of coarse and fine sediments downstream of the dam. J. Hydrol. 615, 128654.

Yao, T., Thompson, L., Yang, W., et al., 2012. Different glacier status with atmospheric circulations in Tibetan Plateau and surroundings. Nat. Clim. Change 2, 663–667.

Zhang, F., Shi, X., Zeng, C., Wang, L., Xiao, X., Wang, G., Chen, Y., Zhang, H., Lu, X., Immerzeel, W., 2020. Recent stepwise sediment flux increase with climate change in the Tuotuo River in the central Tibetan Plateau. Sci. Bull. 65, 410–418.

Zhang, T., Li, D., East, A.E., Kettner, A.J., Best, J., Ni, J., Lu, X., 2023. Shifted sediment-transport regimes by climate change and amplified hydrological variability in cryosphere-fed rivers. Sci. Adv. 9, eadi5019.

Zhang, W., Yang, Y., Zhang, M., Li, Y., Zhu, L., You, X., Wang, D., Xu, J., 2017. Mechanisms of suspended sediment restoration and bed level compensation in downstream reaches of the Three Gorges Projects (TGP). J. Geogr. Sci. 27, 463–480.